

- 13** An oscillator moves with SHM and has a time period of 2.34 s.

How far must it be displaced from its equilibrium position in order that its acceleration is 1.00 ms^{-2} ?

- 14** During SHM a mass moves with an acceleration of 3.4 ms^{-2} when its displacement is 4.0 cm. Calculate its:

a angular frequency **b** time period.

- 15** A mass oscillating on a spring performs exactly 20 oscillations in 15.8 s.

a Determine its acceleration when it is displaced 62.3 mm from its equilibrium position.
b State any assumption that you made when answering **a**.

- 16** Look at the graph in Figure C1.24 which shows the motion of a mass oscillating on a spring. Determine:

a the amplitude
b the time period
c the displacement after 0.15 s
d the displacement after 1.4 s.

- 17 a** Sketch a displacement–time graph showing two complete oscillations for a simple harmonic oscillator which has a time period of 2.0 s and an amplitude of 5.0 cm.

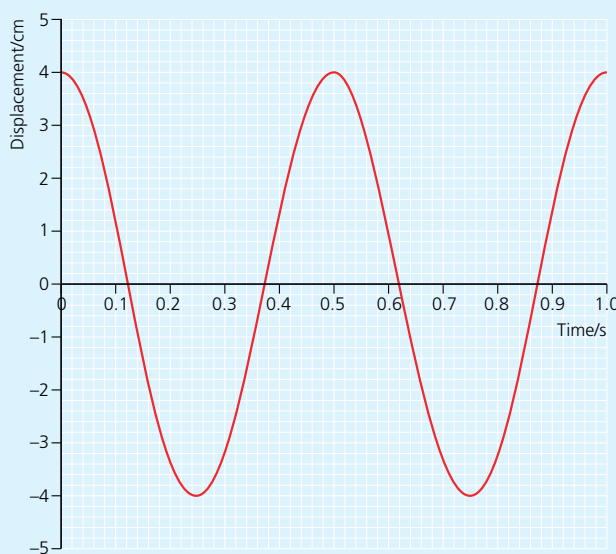
b Add to the same axes the graph of an oscillator which has twice the frequency and the same amplitude.

c Add to the same axes the graph of an oscillator that has an amplitude of 2.5 cm and the same frequency but which is $\frac{1}{4}$ of an oscillation out of phase with the first oscillator.

- 18 a** Sketch a velocity–time graph showing two complete oscillations for a simple harmonic oscillator which has a frequency of 4 Hz and a maximum speed of 4.0 cms^{-1} .

b On the same axes sketch graphs to show the variation of:

i displacement **ii** acceleration
 for the same oscillation.



■ **Figure C1.24** Motion of a mass oscillating on a spring

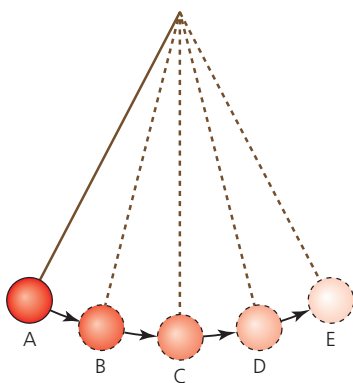
There is more about phase difference, graphs of motion, and the equations which represent them, towards the end of this topic (for HL students).

■ Energy changes during SHM

SYLLABUS CONTENT

- A qualitative approach to energy changes during one cycle of an oscillation.

When an object which can oscillate is pushed, or pulled, away from its equilibrium position, against the action of a restoring force, work will be done and potential energy will be stored in the oscillator. For example, a spring will store elastic potential energy and a simple pendulum will store gravitational potential energy. When the object is released, it will gain kinetic energy and lose potential energy as the restoring force accelerates it back towards the equilibrium position. Its kinetic energy has a maximum value as it passes through the equilibrium position and, at the same time, its potential energy is minimized. As it moves away from the equilibrium position, kinetic energy decreases as the restoring force opposes its motion and potential energy increases again.



■ **Figure C1.25** A swinging pendulum

As an example, consider Figure C1.25, in which a simple pendulum has been pulled away from its equilibrium position, C, to position A. While it is held in position A, it has zero kinetic energy and its change of gravitational potential energy (compared to position C) is greatest. When it is released, gravity provides the restoring force and the pendulum exchanges potential energy for kinetic energy as it moves through position B to position C. At C it has maximum kinetic energy and the change in potential energy has reduced to zero. The pendulum then transfers its kinetic energy back to potential energy as it moves through position D to position E. At E, like A, it has zero kinetic energy and a maximum change of potential energy. The process then repeats every half time period.

If the pendulum was a perfect simple harmonic oscillator, there would be no energy losses, so that the sum of the potential energy and the kinetic energy would be constant and the pendulum would continue to reach the same maximum vertical height and maximum speed every oscillation. In practice, frictional forces will result in energy dissipation and all the energies of the pendulum will progressively decrease.

All mechanical oscillations involve a continuous exchange of energy between kinetic energy and some form of potential energy.

For a simple pendulum, the potential energy E_p is in the form of gravitational potential energy (see Topic A.3 where ΔE_p was used instead of E_p):

$$E_p = mg\Delta h$$

For a mass on a spring, the potential energy is in the form of elastic potential energy (see Topic A.3, where the symbol E_{H} was used instead of E_p and Δx was used instead of x):

$$E_p = \frac{1}{2}kx^2$$

Its maximum value is:

$$E_{p_{\text{max}}} = \frac{1}{2}kx_0^2$$

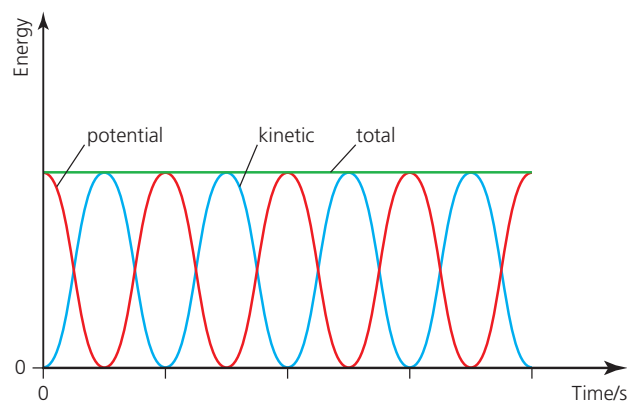
which shows us that generally:

The total energy of an SHM is proportional to its amplitude squared.

Kinetic energy of the mass can be determined from (Topic A.3):

$$E_k = \frac{1}{2}mv^2$$

Figure C1.26 represents these exchanges during several oscillations of perfect SHM.

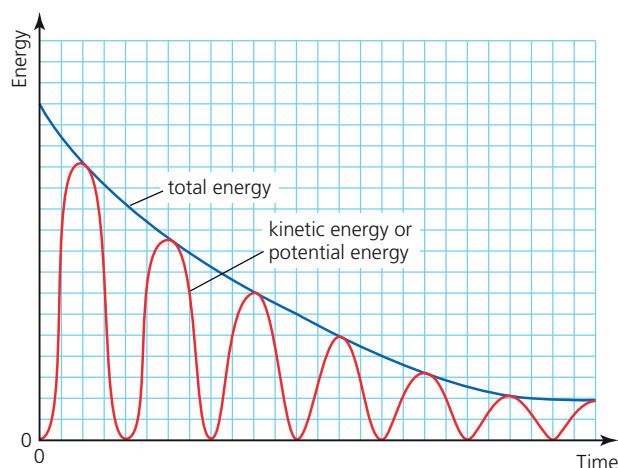


■ **Figure C1.26** Energy changes during oscillation of perfect SHM

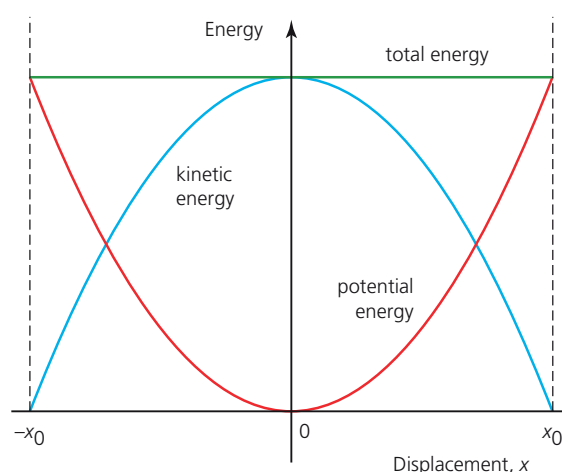
◆ **Damping** Occurs when resistive forces act on an oscillating system, dissipating energy and reducing amplitude.

In more realistic situations, the energies of the system will decrease over time. This is shown in Figure C1.27. Energy dissipation from an oscillating system is called **damping**.

Figure C1.28 shows the variations in energy with displacement during each oscillation of SHM.



■ **Figure C1.27** Energy dissipation in a practical oscillator



■ **Figure C1.28** Variation of energies of a simple harmonic oscillator with displacement

WORKED EXAMPLE C1.5

A mass oscillating in a mass–spring system has a maximum kinetic energy of 0.047 J.

- If its maximum speed was 85 cm s^{-1} , determine its mass.
- State the maximum value of its potential energy.
- Determine the spring constant if the amplitude of the oscillation was 1.9 cm.

Answer

$$\begin{aligned} \text{a } E_k &= \frac{1}{2}mv^2 \\ 0.047 &= \frac{1}{2} \times m \times 0.85^2 \\ m &= 0.13 \text{ kg} \\ \text{b } 0.047 \text{ J} \\ \text{c } E_{p_{\max}} &= \frac{1}{2}kx_0^2 \\ 0.047 &= \frac{1}{2} \times k \times (1.9 \times 10^{-2})^2 \\ k &= 2.6 \times 10^2 \text{ N m}^{-1} \end{aligned}$$

- 19** A simple pendulum, of mass 100 g, is released from rest at its maximum displacement, which is 5.0 cm higher than its central position. It then swings with SHM and a frequency of 1.25 Hz.

- Calculate its maximum potential energy.
- Sketch a graph with fully labelled axes to show the variations in potential energy of the pendulum in the first 1.6 s after it was released.
- Determine the maximum speed of the pendulum.

- 20** The total energy seen in Figure C1.27 shows a nearly 90% decrease in six oscillations.

- Describe where this energy has been transferred to.
- If the oscillating mass was passing through its equilibrium position at time $t = 0$, state what kind of energy is represented by the red line.
- It is suggested that the total energy decreases to the same fraction each oscillation. Analyse data from the graph to check if this is true.